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United States Patent	12415654
Kind Code	B2
Date of Patent	September 16, 2025
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Circumference closing system

Abstract

Various example garment, footwear and bag articles may include a closable circumference in the form of a mouth, waist or foot opening. A circumference closing system may utilize at least one constrict-able tubular length to releasably retain a selected size of the closable circumference or opening.

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Appl. No.: 18/596257

Filed: March 05, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240206596 A1	Jun. 27, 2024

Related U.S. Application Data

continuation parent-doc US 17482859 20210923 US 11930891 child-doc US 18596257
continuation parent-doc US 15654533 20170719 US 11147328 20211021 child-doc US 17482859

Publication Classification

Int. Cl.: B65D33/28 (20060101); A41F9/02 (20060101); A41F1/00 (20060101)

U.S. Cl.:

CPC **B65D33/28** (20130101); **A41F9/025** (20130101); A41F1/00 (20130101); A41F9/02 (20130101)

Field of Classification Search

CPC: B65D (33/28); A41F (1/00); A41F (9/02); A41F (9/025)

USPC: 383/71; 383/72; 383/75; 24/30.5 R

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application claiming priority under 35 USC § 120 from U.S. patent application Ser. No. 17/482,859 filed on Sep. 23, 2021, which is a continuation application of U.S. patent application Ser. No. 15/654,533 filed on Jul. 19, 2017, now U.S. Pat. No. 11,147,328, the full disclosures of which are hereby incorporated by reference.

BACKGROUND

(1) Many pieces of apparel or garments are worn about a person's waist and have a closable waist, a waist that may be constricted to adapt to individual waist sizes as well as personal preferences. Other garments extend over a person's torso and can include a constrictable bottom opening or mid-section, such as draw strings. Such garments come in a variety of different forms such as running shorts, swim shorts, sweatpants, slacks, skirts, bibs, jackets, coats, vests, hoodies, pullovers, sweatshirts, shirts, and the like. Footwear articles have a variety of different lacing or closing systems. Various mechanisms are currently used to control the constriction of the closable waist or the foot opening of footwear, such as a belt, hook and loop fasteners, tri-glides, webbings, snaps,

clips, laces and the like. Such mechanisms may fail, slide or loosen when unloaded or slide or loosen under load. Moreover, such mechanisms may add undesirable hardware, may be weak, may add undesirable amounts of weight, may limit movement, may be difficult to use, or may increase the bulk of the closable waist of the garment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a fragmentary top view of a portion of an example garment that incorporates an example circumference closing system.
- (2) FIG. 2 is a front view of the example garment of FIG. 1.
- (3) FIG. 3 is a front perspective view of another example garment incorporating the example circumference closing system.
- (4) FIGS. 4a and 4b are front perspective views of other example garments incorporating the example circumference closing system.
- (5) FIGS. 5a through 5d are fragmentary top views of portions of example garments incorporating other example implementations of circumference closing systems.
- (6) FIGS. 6a through 6d are fragmentary top views of portions of example garments incorporating other example implementations of circumference closing systems.
- (7) FIG. 7 is a fragmentary front view of a portion of another example garment incorporating another example circumference closing system.
- (8) FIG. 8 is a fragmentary view of the portion of the example
- (9) garment with portions transparently illustrated are shown in broken lines to further illustrate the example circumference closing system.
- (10) FIG. 9 is a top perspective view of the example circumference closing system separated from a remainder of the garment.
- (11) FIG. 10 is an enlarged view of a portion of the circumference closing system of FIG. 9.
- (12) FIG. 11 is an enlarged view of another portion of the circumference closing system of FIG. 9.
- (13) FIG. 12 is an enlarged view of another portion of the circumference closing system of FIG. 9.
- (14) FIG. 13 is a perspective view of an example bag incorporating the example circumference closing system of FIG. 9.
- (15) FIGS. 14-17 are front views illustrating an example method for forming a portion of the example circumference closing system of FIG. 9.
- (16) FIG. 18 is a top view of an article of footwear incorporating an example implementation of a footwear closing system.
- (17) FIGS. 19 and 20 are top views of articles of footwear incorporating other example implementations of footwear closing systems.

DETAILED DESCRIPTION OF EXAMPLES

(18) Disclosed herein are example garments that provide a closable circumference or waist and that utilize a circumference closing system that utilizes at least one constrict-able tubular length to releasably retain a selected size of the closable circumference. The disclosed circumference closing system may be used to supplement or replace existing mechanisms or hardware in a variety of garments and other applications. The disclosed circumference closing system may eliminate undesirable hardware, may reduce the weight of the garment, may increase movement or flexibility of the garment or may reduce the bulk of the closable waist of the garment.

(19) FIGS. 1 and 2 illustrate an example garment 20, shown as a generic pair of pants. FIG. 1 is a top view of a portion of pants 20 while FIG. 2 is a front view of pants 20. Pants 20 may have various forms and configurations and go by various name other names such as slacks, sweatpants, trousers, tights, jeans, gauchos, pedal pushers, palazzos, pajamas, jodhpurs, knickers, bibs and the

like.

(20) As shown by FIG. 2, pants **20** generally comprises waist portion **22**, hips portion **24**, crotch **26**, leg portions **28** and circumference closing system **30**. Waist portion **22** comprise a body of material, such as woven material providing a constricted or constrict-able opening **34** that receives a person's waist and that supports the remaining portions of pants **20** when being worn by a person. As shown by FIG. 1, in the example illustrated, waist portion **22** has an internal sleeve **36** that receives portions of circumference closing system **30**.

(21) In one implementation, waist portion **22** provides a continuous uninterrupted body of material that continuously extends about a person's waist. One example of such an implementation is pants having an uninterrupted body of material that continuously extends about a person's waist is a sweatpants. In another implementation, waist portion **22** may be split or have a front zipper, wherein opposite sides of the split are temporarily clasped, buttoned or otherwise secured to one another. In other examples, pants can include one or more side zipper or reclosable slits or splits. In all such cases, waist portion **22** is flexible and/or compressible so as to be constrict-able, facilitating adjustment of the size of waist portion **22** through the use of circumference closing system **30**.

(22) Hips portion **24** comprise that portion of pants **20** extending from waist portion **22** to crotch **26**. Hips portion **24** extends about a person's hips when pants **20** is worn. In some implementations, hips portion **24** may have at least one pocket secured thereto on the front or sides.

(23) Leg portions **28** extend from hips portion **24**, beginning at crotch **26** and terminating at leg openings **38**. The distance between crotch **26** and leg openings **38** is sometimes referred to as an inseam. Although illustrated as substantially straight, leg portions **26** may have a variety of shapes and configurations such as loose fit, tapered, bell bottom and the like.

(24) Circumference closing system **30** engages waist portion **22** and is closable or constrict-able so as to adjust the circumference or the interior size of waist portion **22** to accommodate different size waists and/or other undergarments of users who may be wearing garment **20**. Circumference closing system **30** comprises a single line **40** at least partially received within sleeve **36**. Line **40** comprises portions **42**, **44**, **46**, **48** and **50**.

(25) Portion **42** comprises a first end portion of line **40** secured to waist portion **22**. In one implementation, portion **42** is stitched to waist portion **22**. In another implementation, portion **42** is welded, bonded or otherwise secured to waist portion **22**.

(26) Portion **44** of line **40** extends from portion **42** and continues along waist portion **22**, within sleeve **36**, about substantially an entirety of the closable circumference of waist portion **22**. In the example illustrated, portion **44** comprises a constrict-able tubular length **50** and an exposed portion **52**. Constrict-able tubular length **50** comprises a length of line **40** which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length **50**. In one implementation, the entire length of line **40** is a constrict-able tube. For example, in one implementation, the entire length of line **40** comprises a braided tube of material. In another implementation, the entire length of lines **40** comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length **50** may comprise a portion of line **40**, wherein other portions of line **40** are not tubular or are not constrict-able.

(27) Portion **46** comprises that portion of line **40** that is extending through the hollow interior of constrict-able tubular length **50**, entering at entry **54** and exiting at exit **56**. Portion **46** is sized to enable portion **46** to pass through the constrict-able tubular length **50**. Portion **48** comprises that portion of line **40** that extends from exit **56** and from length **50**. Portion **48** provides a segment of line **40** by which a person may grasp and pull line **40** to adjust a length of line **40** that extends from portion **42**, about the waist of the person wearing garment **20**, and to the constriction of length **50**. In one implementation, portion **48** is a loose, unattached second end of line **40**. In another implementation, an end of portion **48** may be secured to waist portion **22** so as to not hang or

dangle.

(28) Exposed portion **52** comprises a segment of portion **44** of line **40** that is exposed outside of waist portion **22** and sleeve **36** such that a person may grasp portion **52** to pull back portion **48** through length **50** (as indicated by arrow **58**), shortening the exposed length of portion **48** and increasing the length of portion **44** extending between the constriction of length **50**, about the waist of the person wearing garment **20** and back to portion **42**. In other words, exposed portion **52** facilitates lengthening of portion **44** and widening of the closable circumference of waist portion **22**.

(29) In the example illustrated, sleeve portion **36** and waist portion **22** comprise openings **64**, **66** and **68**. In one implementation, each of opening **64**, **66** and **68** may be defined or protected by a grommet or other lining. Opening **64** comprise an opening from which portion **44** exits sleeve **36** and waist portion **22**. Opening **66** comprises an opening into which portion **44** reenters sleeve **36** and waist portion **44**, further extending towards constrict-able tubular length **50**. Openings **64** and **66** facilitate the provision of exposed portion **52**, allowing line **44** to be loosened when the size of waist portion **22** is to be increased. Opening **68** comprise an opening in sleeve **36** and waist portion **22** through which portion **48** exits, facilitating grasping a portion **48** when waist portion **22** is be constricted, closed or reduced in size.

(30) When a person wishes to reduce or constrict the size of closable circumference of waist portion **42**, the person may pull portion **48** through length **50** (as indicated by arrow **60**), increasing the length of portion **48** and reducing the length of portion **44** extending between the constriction of length **50**, about the waist of the person wearing garment **20** in back to portion **42**. After the length of portion **44** in the size of the closable circumference of waist portion **22** have been established, circumference closing system **30** retains the chosen size of the closable circumference of waist portion **42**. Outward forces exerted upon waist portion **42** and upon line **44** place an axial load on length **50** which constricts about and against portion **46**. The constriction of portion **50** about portion **46** frictionally locks portion **50** about the entire circumference or about all sides of portion **46**, inhibiting axial sliding movement of portion **46** within the hollow interior of constrict-able tubular length **50**.

(31) As shown by FIG. **1**, circumference closing system **30** provides garment **20** with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system **30** provides garment **20** with an easily adjustable waist portion **22** without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment **20**. Although circumference closing system **30** is illustrated as being part of pants **20**, circumference closing system **30** may be used in waist portions of other pieces of apparel or other garments.

(32) FIG. **3** illustrates another example piece of apparel or garment **120** which comprises circumference closing system **30**. Garment **120** is in the form of a pair of shorts. For purposes of this disclosure, the term “shorts” refers to any garment worn about a person's waist having shorter leg portions than the leg portions **28** of pants **20** or no leg portions. “Shorts” also refers to any garment about a person's waist that have leg openings leg openings that are to be located above the person's ankles, above a person's knees or in some implementations, reach only to a person's thighs. The term “shorts” comprises all forms of shorts including hiking shorts, casual shorts, compression shorts, short tights, swim trunks, capris, skorts and culottes.

(33) As shown by FIG. **3** garment **120** comprises waist portion **22**, hips portion **24**, crotch portion **26**, leg portions **128** and circumference closing system **30**. Waist portion **22**, hips portion **24**, crotch portion **26** and circumference closing system **30** are described above. Leg portions **128** are similar to leg portions **28** except that leg portions **128** are shorter. In the example illustrated, leg portions **128** are sized so as to provide leg openings **38** which are to be located between the thighs and knees of a person wearing garment **120**. In other implementations, leg portions **128** may be sized so as to

provide leg opening **38** closer to crotch **26** or below a person's knees when garment **120** is worn. In other implementations, one or more of the leg portions **28** or one or more leg openings **38** may include circumference closing system **30**.

(34) FIG. **4a** illustrates another example piece of apparel or garment **220** which comprises circumference closing system **30**. Garment **220** is in the form of a skirt. For purposes of this disclosure, the term “skirt” refers to any garment worn about a person's waist and generally lacking a crotch or inseam. Examples of different types of skirts include, but not limited to, A-line skirts, broomstick skirts, bubble skirts, circular skirts, dirndl skirt, fishtail skirts, flared skirts, gored skirts, fly skirts, sheath, multi-panel skirts, box pleated skirts, knife pleats skirts, fix-box pleated skirts, prairie skirts, kilts, wrap skirts, jean skirts, tiered skirts, pencil skirts, sarongs, pareos, tulle skirts, handkerchief skirts, trumpet skirts, round skirts, wrap skirts, bazi skirt, cowl skirts, godet skirts and the like.

(35) As shown by FIG. **4a**, garment **220** comprises waist portion **22**, skirt body **124** and circumference closing system **30**. Waist portion **22** and circumference closing system **30** are described above. Skirt body **124** hangs from waist portion **22**. As described above, skirt body **124** may be provided in different sizes, shapes and configurations and formed from a single panel or multiple panels of material.

(36) FIG. **4b** illustrates another example piece of apparel or garment **270** which comprises circumference closing system **30**. Garment **270** is in the form of a pullover jacket. For purposes of this disclosure, the term “jacket” refers to any garment worn about a person's torso including coats, vests, hoodies, pullovers (which are pulled over a person's head) and the like. Although FIG. **4b** illustrates a pullover jacket, the closing system **30** can be applied to any jacket. The jacket includes a neck region **280**, shoulder regions **282**, a torso region **286** and arms **288**, and an open lower end **272**. A constrict-able region **274** can be positioned at the lower end **272** and/or at other locations about jacket. The arms **288** include arm openings **290**. For example, a longer jacket can extend below a user's waist, and may include a first constrict-able region **274** at the waist of the user and/or a second constrict-able region **274** at the open lower end **272** below the waist of the user. Garment **270** includes the circumference closing system at the constrict-able region **274** for selectably adjusting the size of the open lower end and/or the constrict-able region. In other implementations, one or more of the arm openings **290** may include circumference closing system **30**.

(37) FIG. **5a** illustrates another example garment **320**. Garment **320** may have the general form of any of garments **20**, **120**, **220** and **270** described above except that garment **320** comprises circumference closing system **330**. Garment **320** comprises waist portion **22** of a closed system having a continuous waist portion **22** without a separation. In one implementation, waist portion **22** of garment **330** comprises a sleeve **36**. In other implementations, sleeve **36** may be omitted. Although circumference closing system **330** is illustrated as being largely contained within waist portion **22**, in other implementations, larger portions if not the entirety of circumference closing system **330** may be exposed or external to waist portion **22**.

(38) Circumference closing system **330** operates in a somewhat similar fashion to circumference closing system **30** except that circumference closing system **330** utilizes a pair of loops and a pair of lines, wherein adjustment of waist portion **22** occurs through the pair of loops. As shown by FIG. **5a**, system **330** comprises loops **332** and lines **340A**, **340B** (collectively referred to as line **340**). Loops **332** are located on a front side of garment **320** along waist portion **22** and are connected to portions of waist portion **22** that collectively extend about the remainder of waist portion **22**. In the example illustrated, circumference closing system **330** additionally comprises a band **334** extending within sleeve **36** which has opposite ends secured to loops **332** and extends about the remainder of the interior circumference **34** of waist portion **22**. In other implementations, band **334** may be omitted where loops **332** are affixed to other portions of waist portion **22** or other locations of a garment.

(39) Lines **340** facilitate selected drawing of loops **332** towards one another or the release of such loops **332** to allow movement away from one another so as to thereby adjust the circumference, size or extent of constriction of opening **34** of waist portion **22**. Each of lines **340** is substantially similar or identical to one another except that lines **340A**, **340B** mirror one another on opposite sides of a front centerline of garment **320**. Each of lines **340** comprises portions **342**, **344**, **346**, **348** and **350**.

(40) Portion **342** comprise an end portion of line **340A**, **340B** secured to waist portion **22**. In one implementation, portion **342** is stitched to waist portion **22**. In another implementation, portion **342** is welded, bonded or otherwise secured to waist portion **22**. Referring to FIG. *5b*, in another implementation, the circumference closing system **330** can be applied to a garment **320** having a separation or break **335** between the portions **342**. The separation or break **335** can be a fly of a pant, a full separating jacket hem or other separation.

(41) Referring to FIG. *5a*, portion **344** of lines **340A**, **340B** extends from portion **342** and continues through one of loops **332**, within sleeve **36**, returning towards portion **342**. In the example illustrated, portion **344** comprises a constrict-able tubular length **350** and an exposed portion **352**. In the example illustrated, portion **344** comprises a constrict-able tubular length **350** and an exposed portion **352**. Constrict-able tubular length **350** comprises a length of line **340A**, **340B** which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length **350**. In one implementation, the entire length of line **340A**, **340B** is a constrict-able tube. For example, in one implementation, the entire length of line **340A**, **340B** comprises a braided tube of fibrous material. In another implementation, the entire length of lines **340** comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length **350** may comprise a larger or smaller portion of line **340A**, **340B**, wherein other portions of lines **340A**, **340B** are not tubular or are not constrict-able. In still other implementations the constrict-able tubular length **350** can be positioned at any location between portion **342** and the loop **332**.

(42) Portion **346** comprises that portion of line **340A**, **340B** that is extending through the hollow interior of constrict-able tubular length **350**, entering at entry **354** and exiting at exit **356**. Portion **348** comprises that portion of line **340A**, **340B** that extends from exit **356** and from length **350**. Portion **48** provides a segment of line **340A**, **340B** by which a person may grasp and pull line **340A**, **340B** to adjust a length of line **340A**, **340B** that extends from portion **42**, about the waist of the person wearing garment **320**, and to the constriction of length **350**. In the example illustrated, portion **348** is secured to waist portion **22** at end portion **357**. In other implementations, portions **348** may comprise a loose, unattached end.

(43) Exposed portion **352** comprises a segment of portion **344** of line **340A**, **340B** that is exposed inside of waist portion **22** and sleeve **36** such that a person may grasp portion **352** to pull back portion **346** through length **350** (as indicated by arrows **358**), shortening the length of portion **348** and increasing the length of portion **344** extending from the constriction of length **350**, through loop **332** and back to portion **342**. As the length of portion **348** is pulled back through length **350**, length **350** is in an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when length **350** is in a constricted state), better allowing portion **348** to be pulled back through length **350** towards loop **332**. Exposed portion **352** facilitates lengthening of portion **344** and widening of the closable circumference **34** of waist portion **22**.

(44) In the example illustrated, sleeve portion **36** and waist portion **22** comprise openings **364**, **366**, **368** and **370**. In one implementation, each of opening **364**, **366**, **368** and **370** may be defined or protected by a grommet or other lining. In FIGS. *5a* and *5d*, opening **364** comprises an opening from which portion **344** exits sleeve **36** and waist portion **22**. Opening **366** comprises an opening into which portion **344** reenters sleeve **36** and waist portion **22**, further extending through loop **332** and back towards constrict-able tubular length **350**. Openings **364** and **366** facilitate access to

exposed portion 352, allowing line 344 to be loosened when the size of waist portion 22 is to be increased.

(45) Referring to FIG. 5a, opening 368 comprises an opening in sleeve 36 and waist portion 22 through which portion 348 exits, facilitating grasping of portion 348 when waist portion 22 is to be constricted, closed or reduced in size. Opening 370 allows end portion 357 of portion 348 to extend back into waist portion 22 for securement to waist portion 22.

(46) Referring to FIGS. 5a and 5b, when a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22, the person may pull portion 348 through length 350 (as indicated by arrow 360), increasing the length of portion 348 and reducing the length of portion 340A and/or 340B extending from the constriction of length 350, through loop 332 and back to portion 342. After the length of portion 344 and the size of the closable circumference of waist portion 22 have been established, the constriction of portion 350 retains the chosen size of the closable circumference of waist portion 22. Outward forces exerted upon waist portion 22 and upon line portions 344 of lines 340A, 340B place an axial load on length 350 which constricts about and against portion 346. The constriction of portion 350 about portion 346 frictionally locks portion 350 to portion 346 about the entire circumference or on all sides of portion 346, inhibiting axial sliding movement of portion 346 within the hollow interior of constrict-able tubular length 350.

(47) In other implementations end portion 357 could be secured to portion 344 on the opposite side of opening 354 from constrict-able tubular length 350. In FIG. 5b, by grasping portion 348 and applying force in direction 371 toward portion 342, a person can take the axial load off of portion 350, reducing the constriction on portion 346 and lengthening portion 344 which increases the total circumference of waist opening 330. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect. In other implementations in which end portion 357 of portion 348 is unattached, opening 370 may facilitate stowing of the unattached end portion to extend back into waist portion 22 for securement to waist portion 22, if other means of loosening are applied.

(48) Referring to FIG. 5c in one implementation, portion 348 can be divided into tail end portions 341 and 343. End portion 341 passes through opening 370 allowing end portion 357 of portion 341 to extend back into waist portion 22 for securement to waist portion 22. In other implementations end portion 357 could be secured to portion 344 on the opposite side of opening 354 from constrict-able tubular length 350. When portion 348 is divided into tail portions 341 and 343, the tail portion 341 can include a graspable end 359 and the portion 343 can include a graspable end 349. Similar to the implementations of FIGS. 5a and 5b, when a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5c, the person may pull end portion 343 through length 350 (as indicated by arrow 360), increasing the length of portion 343 and reducing the length of portion 340 extending from the constriction of length 350, through loop 332 and back to portion 342. When a person wishes to increase the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5c, a person can grasp end portion 341 and apply force in direction 371 toward 342, which takes the axial load off of portion 350, reduces the constriction on portion 346, and lengthens portion 344 which increases the total circumference of waist opening 34. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

(49) Referring to FIG. 5d, another implementation of the present invention is illustrated with portion 352 providing access for a person to increase the size of closable circumference 34 of waist portion 22, and includes portion 348 as an end portion. When a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5d, the person may pull end portion 348 through length 350 (as indicated by arrow 360), increasing the length of portion 348 and reducing the length of portion 340A extending from the constriction of length 350, through loop 332 and back to portion 342. When a person wishes to increase the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5d, a person can grasp

portion **352** apply force in direction of arrow **358**, which takes the axial load off of portion **350**, reduces the constriction on portion **346**, and lengthens portion **344** which increases the total circumference of waist opening **34**. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

(50) As shown by FIGS. **5a** through **5d**, circumference closing system **330** provides garment **320** with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system **330** provides garment **320** with an easily adjustable waist portion **22** without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment **320**. Although circumference closing system **330** is illustrated as being part of pants **20**, circumference closing system **30** may be used in other pieces of apparel or other garments. In another implementation, the circumference closing system **330** can be configured with only a single loop **332**. In another implementation, the circumference closing system **330** can be configured with only a single line **340A** or **340B** or with a line **340A** or **340B** is reduced in size. For example, one of the ends **357** can be directly connected to one of the ends of the band **334** thereby eliminating one of the loops **340A** or **340B** and/or one of the loops **332**.

(51) FIGS. **6a** through **6d** illustrate another example garments **420**. Garment **420** may have the general form of any of garments **20**, **120**, **220** and **270** described above except that garment **420** comprises circumference closing system **430**. Garment **420** comprises waist portion **22**. In one implementation, waist portion **22** of garment **420** comprises a sleeve **36**. In other implementations, sleeve **36** may be omitted. Although circumference closing system **430** is illustrated as being largely contained within waist portion **22**, in other implementations, larger portions if not the entirety of circumference closing system **430** may be exposed or external to waist portion **22**.

(52) Circumference closing system **430** operates in a somewhat similar fashion to circumference closing systems **30** and **330** described above except that circumference closing system **430** utilizes interconnected lines to adjust waist portion **22**. As shown by FIG. **6a**, system **430** comprises lines **440A**, **440B** (collectively referred to as lines **440**).

(53) Lines **440** facilitate adjustment of the circumference, size or extent of constriction of opening **34** of waist portion **22**. Each of lines **440** is substantially similar or identical to one another except that lines **440A**, **440B** mirror one another on opposite sides of a front centerline of garment **420**. Line **440A** comprises portions **442A**, **444A**, **446A** and **448A**.

(54) Portion **442A** comprise an end portion of line **440A** secured to waist portion **22**. In one implementation, portion **442A** is stitched to waist portion **22**. In another implementation, portion **442A** is welded, bonded or otherwise secured to waist portion **22**.

(55) Portion **444A** of line **440A** extends from portion **442** within sleeve **36** and further extends substantially about the entire circumference **34** of waist portion **22**. In the example illustrated, portion **444A** comprises a constrict-able tubular length **450A** and an exposed portion **452A**. In the example illustrated, portion **444A** comprises a constrict-able tubular length **450A** and an exposed portion **452A**. Constrict-able tubular length **450A** comprises a length of line **440A** which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length **450A**. In one implementation, the entire length of line **440A** is a constrict-able tube. For example, in one implementation, the entire length of line **440A** comprises a braided tube of material. In another implementation, the entire length of line **440A** comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length **450A** may comprise a portion of line **440A**, wherein other portions of line **440A** are not tubular or are not constrict-able.

(56) Portion **446A** comprises that portion of line **440A** that is extending through the hollow interior of constrict-able tubular length **450B**, entering at entry **454A** and exiting at exit **456A**. Portion **448** comprises that portion of line **440A** that extends from exit **456A** and from length **450B**. Portion

448A provides a segment of line **440A** by which a person may grasp and pull line **440A** to adjust a length of line **440A** that extends from portion **442A**, about the waist of the person wearing garment **420**, and through the constriction of length **450B**. In the example illustrated, portion **448A** is secured to waist portion **22** at end portion **457A** so as to not hang or dangle. In other implementations, portions **448A** may comprise a loose, unattached end.

(57) Exposed portion **452A** comprises a segment of portion **444A** of line **440A** that is exposed outside of waist portion **22** and sleeve **36** such that a person may grasp portion **452A** to pull back portion **448A** through length **450B** (as indicated by arrow **458A**), shortening the length of portion **448A** and increasing the length of portion **444A** extending from the constriction of length **450B**. As the length of portion **448A** is pulled back through length **450B**, length **450A** releases the axial load on length **450B**, returning it to an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when length **450B** is in a constricted state), better allowing portion **448B** to be pulled back through length **450B**. Exposed portion **452A** facilitates lengthening of portion **444A** and widening of the closable circumference **34** of waist portion **22**. The constricting and release of the closing system **430** can be formed on one or both sides of garment **420** to achieve the desired tightening and/or loosening of the waist portion **22**.

(58) Line **440B** is similar to line **440A**. Line **440B** comprises portions **442B**, **444B**, **446B** and **448B**. Portion **442B** comprise an end portion of line **440B** secured to waist portion **22**. In one implementation, portion **442B** is stitched to waist portion **22**. In another implementation, portion **442B** is welded, bonded or otherwise secured to waist portion **22**.

(59) Portion **444B** of line **440B** extends from portion **442B** within sleeve **36** and further extends substantially about the entire circumference **34** of waist portion **22**. In the example illustrated, portion **444B** comprises the constrict-able tubular length **450B** (noted above) and an exposed portion **452B**. Constrict-able tubular length **450B** comprises a length of line **440B** which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length **450B**. In one implementation, the entire length of line **440B** is a constrict-able tube. For example, in one implementation, the entire length of line **440B** comprises a braided tube of fibrous material such as nylon or polyester. In another implementation, the entire length of line **440B** comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length **450B** may comprise a shorter or longer portion of line **440B**, wherein other portions of line **440B** are not tubular or are not constrict-able.

(60) Portion **446B** comprises that portion of line **440B** that is extending through the hollow interior of constrict-able tubular length **450A**, entering at entry **454B** and exiting at exit **456B**. Portion **448B** comprises that portion of line **440B** that extends from exit **456B** and from length **450A**. Portion **448B** provides a segment of line **440B** by which a person may grasp and pull line **440B** to adjust a length of line **440B** that extends from portion **442B**, about the waist of the person wearing garment **420**, and to the constriction of length **450A**. In the example illustrated, portion **448B** is secured to waist portion **22** at end portion **457B** so as to not hang or dangle. In other implementations, portions **448B** may comprise a loose, unattached end.

(61) Exposed portion **452B** comprises a segment of portion **444B** of line **440B** that is exposed inside of waist portion **22** and sleeve **36** such that a person may grasp portion **452B** to pull back portion **448B** through length **450A** (as indicated by arrow **458B**), shortening the length of portion **448B** and increasing the length of portion **444B** extending from the constriction of length **450A**. As the length of portion **448B** is pulled back through length **450A**, length **450B** releases the axial load on length **450A**, returning it to an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when length **450A** is in a constricted state), better allowing portion **448B** to be pulled back through length **450A**. Exposed portion **452B** facilitates lengthening of portion **444B** and widening of the closable circumference **34** of waist portion **22**.

(62) In the example illustrated, sleeve portion **36** and waist portion **22** comprise openings **464A**,

464B, **466A**, **466B**, **468A**, **468B** and **470A**, **470B** (collectively referred to as openings **464**, **466**, **468** and **470**, respectively). In one implementation, each of openings **464**, **466**, **468** and **470** may be defined or protected by a grommet or other lining. Each of openings **464** and **466** comprises an opening from which portion **444A**, **444B** exits sleeve **36** and waist portion **22**. Each of openings **466** comprises an opening into which portion **444A**, **444B** reenters sleeve **36** and waist portion **22**. Openings **446** facilitate the provision of exposed portions **452A**, **452B**, allowing lines **444A**, **444B**, respectively, to be loosened when the size of waist portion **22** is to be increased. Each of openings **468** comprises an opening in sleeve **36** and waist portion **22** through which portion **448A**, **448B** exits, facilitating grasping of portion **448A**, **448B** when waist portion **22** is to be constricted, closed or reduced in size. Each of openings **470** allows the option for end portions **457A**, **457B** of portions **448A**, **448B**, respectively, to extend back into waist portion **22** for securement to waist portion **22**. In other implementations in which end portions **457** of portions **448** is unattached, opening **470** may facilitate stowing of the unattached end portion.

(63) When a person wishes to reduce or constrict the size of closable circumference **34** of waist portion **22**, the person may pull on both portions **448**. For example, the person may pull portion **448A** through length **450B** (as indicated by arrow **360A**), increasing the length of portion **448A** and reducing the length of portion **444A** extending about waist portion **22** and back to portion **442A**. The person may simultaneously pull portion **448B** through length **450A** (as indicated by arrow **360B**), increasing the length of portion **448B** and reducing the length of portion **444B** extending about waist portion **22** and back to portion **442A**. After the length of portions **444A**, **444B** and the size of the closable circumference of waist portion **22** have been established, the constriction of portions **450A** and **450B** retains the chosen size of the closable circumference of waist portion **22**. Outward forces exerted upon waist portion **22** and upon line portions **444A**, **444B** of lines **440A**, **440B** place an axial load on lengths **450A**, **450B** which constricts about and against portions **446A**, **446B**, respectively. The constriction of portions **450** about portions **446** frictionally locks portions **450** to portions **446** about the entire circumference or in all sides of portions **446**, inhibiting axial sliding movement of portions **446** within the hollow interior of constrict-able tubular lengths **450**.

(64) Referring to FIG. **6b**, in another implementation, the circumference closing system **420** can be applied to a garment **420** having a separation or break **435** between the portions **442A** and **B**. The separation or break **435** can be a fly of a pant, a full separating jacket hem or other separation.

(65) Referring to FIGS. **6a** and **6b**, when a person wishes to reduce or constrict the size of closable circumference **34** of waist portion **22**, the person may pull portions **448A** and **B** through lengths **450A** and **B** (as indicated by arrows **460A** and **B**), increasing the length of portions **448A** and **B**. After the desired lengths of portions **444A** and **B** and the size of the closable circumference of waist portion **22**, have been established, the constriction of portions **450A** and **B** retain the selected size of the closable circumference of waist portion **22**. Outward forces exerted upon waist portion **22** and upon line portions **444A** and/or **B** place an axial load on lengths **450A** and **B** which constricts about and against portion **446B** and **A**, respectively. The constriction of portions **450A** and **B** about portions **446B** and **A**, respectively, frictionally locks portion **450A** and **B** to portions **446B** and **A**, respectively, about the entire circumference or on all sides of portions **446A** and **B**, inhibiting axial sliding movement of portion **446B** and **A** within the hollow interior of constrict-able tubular length **450A** and **B**, respectively.

(66) In FIG. **6b**, by grasping portion **448A** and **B**, and applying force in direction **471** toward **442A** and **B**, a person can take the axial load off of portions **450B** and **A**, respectively, reducing the constriction on portions **446A** and **B** and lengthening portions **444A** and **B**, which increases the total circumference of waist opening **430**. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

(67) Referring to FIG. **6c** in one implementation, portion **441A** passes through opening **470A** allowing end portion **457A** of portion **444A** for securement to waist portion **22**. Portion **441A** can include a graspable end **459**. Similarly, portion **448B** can include a graspable end **449**. When a

person wishes to reduce or constrict the size of closable circumference **34** of waist portion **22** of the implementation of FIG. **6c**, the person may pull portion **448B** through length **450A** (as indicated by arrow **460B**), increasing the length of portion **448B** extending from the constriction of length **450A** and reducing the size of the circumference **34**. When a person wishes to increase the size of closable circumference **34** of waist portion **22** of the implementation of FIG. **6c**, a person can grasp end portion **441A** apply force in direction **471** toward portion **442A**, which takes the axial load off of portion **450A**, reduces the constriction on portion **446B**, and lengthens portion **444B** which increases the total circumference of waist opening **34**. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

(68) Referring to FIG. **6d**, another implementation of the present invention is illustrated with portion **452B** providing access for a person to increase the size of closable circumference **34** of waist portion **22**, and includes portion **448B** as an end portion. When a person wishes to reduce or constrict the size of closable circumference **34** of waist portion **22** of the implementation of FIG. **6d**, the person may pull end portion **448B** through length **450A** (as indicated by arrow **460B**), increasing the length of portion **448B** extending from the constriction of length **450A** thereby reducing the length of **444B**. When a person wishes to increase the size of closable circumference **34** of waist portion **22** of the implementation of FIG. **6d**, a person can grasp portion **452B** apply force in direction of arrow **458B**, which takes the axial load off of portion **450A**, reduces the constriction on portion **446B**, and lengthens portion **444B** which increases the total circumference of waist opening **34**. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

(69) As shown by FIGS. **6a** through **6d**, circumference closing system **430** provides garment **420** with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system **430** provides garment **420** with an easily adjustable waist portion **22** without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment **420**. Although circumference closing system **430** is illustrated as being part of pants **20**, circumference closing system **430** may be used in other pieces of apparel or other garments.

(70) FIGS. **7-12** illustrate another example garment **520** comprising a circumference closing system **530**. The example is similar to the implementation of FIG. **6d**. In the example illustrated, garment **520** comprises a pair of pants or shorts having split circumference **534** (separated at a zipper region or split **535**), wherein opposite sides of the split **535** are secured to one another by a clasp or hook **537**. In other implementations, split **535** may be omitted. In other implementations, garment **520** may comprise an alternative piece of apparel or garment, such as a skirt or a jacket.

(71) FIGS. **9-12** illustrate circumference closing system **530** in more detail, separated and removed from the remainder of garment **520**. As shown by FIG. **9**, circumference closing system **530** comprises band **600**, guides **602**, and lines **540A**, **540B** (collectively referred to as lines **540**). Band **600**, guide **602** and lines **540** are substantially contained within the interior of sleeve **536** along waist portion **522**.

(72) Band **600** comprises an elongate strap or webbing that extends within sleeve **536** about the circumference **534** of waist portion **522**. Band **600** as opposite ends affixed to clasp **537**, joining opposite sides of waist portion **522** to form an endless circumference. In the example illustrated, band **600** comprises a strap of elastic material such as polyester or other elastomeric material. The elastic nature of band **600** provide some degree of accommodation for different waist sizes. Moreover, the elastic nature of band **600** distributes loads of lines **540A** and **B** along band **600**. In other implementations, band **600** may be inelastic. In yet other implementations, band **600** may be omitted, where lines **540** are contained within sleeve **36** or extend on the exterior waist portion **522**.

(73) As shown by FIG. **10**, guides **602** comprise channels provided at spaced locations along the length of band **600**. Guides **602** provide control passages through which lines **540** pass. Guides **602**

maintain lines **540** in close proximity and opposite to band **600** such that changes in the “effective” length of lines **540** results in adjustment of the circumference of band **600** and the circumference **534** of waist portion **522**. In other implementations, guide **602** may be omitted or of other configurations.

(74) Line **540A** comprises portions **542A**, **544A**, **546A**, **548A** and **550A**. Portion **542A** comprise an end portion of line **540A** secured to band **600** in one implementation, portion **542A** is stitched to band **600**. In another implementation, portion **542A** is welded, bonded or otherwise secured to band **600**.

(75) Portion **544A** of line **540A** extends from portion **542a** along band **600**, guided by guides **602**, within sleeve **536** and further extends substantially about the entire circumference **534** of waist portion **522**. In the example illustrated, portion **544A** comprises a constrict-able tubular length **550A** (shown in FIGS. **8** and **11**) and an exposed portion **552A** (FIG. **8**). In the example illustrated, portion **544A** comprises a constrict-able tubular length **550A**. Constrict-able tubular length **550A** comprises a length of line **540A** which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length **550A**. In one implementation, the entire length of line **540A** is a constrict-able tube. For example, in one implementation, the entire length of line **540A** comprises a braided tube of material. In another implementation, the entire length of line **540A** comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length **550A** may comprise a portion of line **540A**, wherein other portions of line **540A** are not tubular or are not constrict-able.

(76) Referring to FIGS. **11** and **12**, portion **546A** comprises that portion of line **540A** that is extending through the hollow interior of constrict-able tubular length **550B**, entering at entry **554** and exiting at exit **556A** where line **540A** passes through band **600**, forming portion **548A**. Portion **548A** comprises that portion of line **540A** that extends from exit **556A** and from length **550B**. Portion **548A** provides a segment of line **540A** by which a person may grasp and pull line **540A** to adjust a length of line **540A** that extends from portion **542A**, about the waist of the person wearing garment **520**, and to the constriction of length **550B**. In the example illustrated, portion **548A** comprises a loose, unattached end. In other implementations, portion **548A** is secured to waist portion **522** so as to not hang or dangle, or to reattach portion **548A** to portion **544** adjacent to entry **554** opposing length **550** in order to loosen.

(77) Line **540B** is similar to line **540A** except that line **540B** has portion **542B** that has an exposed portion **552B** (shown in FIG. **8**). Those remaining components of line **540B** which correspond to components of line **540A** are numbered similarly but followed with a “B” designation rather than an “A” designation.

(78) Exposed portion **552B** comprises a segment of portion **544B** of line **540A** that is exposed inside of waist portion **522** and sleeve **536** such that a person may grasp portion **552B** to pull back portion **548B** through length **550A** (as indicated by arrow **558B**), shortening the length of portion **548B** and increasing the length of portion **544B** extending from the constriction of length **550A**. As the length of portion **548B** is pulled back through length **550A**, tension on length **550B** is reduced such that both of lengths **550A** and **550B** are in an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when in a constricted state), better allowing portion **548A** to be pulled back through length **550B** when a load is applied at **537A**. Exposed portion **552B** facilitates the widening of the closable circumference **534** of waist portion **522**.

(79) In the example illustrated by FIGS. **7** and **8**, sleeve portion **536** and waist portion **522** comprise openings **564A**, **564B**, **566A**, **566B** and **568A**, **568B** (collectively referred to as openings **564**, **566** and **568**, respectively). In one implementation, each of openings **564**, **566** and **568** may be defined or protected by a grommet or other lining. Each of openings **564** comprises an opening from which portion **544A**, **544B** exits sleeve **536** and waist portion **522**. Each of openings **566** comprises an opening into which portion **544A**, **544B** reenters sleeve **536** and waist portion **522**.

Openings **564** facilitate the provision of exposed portions **552A**, **552B**, allowing lines **544A**, **544B**, respectively, to be loosened when the size of waist portion **522** is to be increased. Each of openings **568** comprises an opening in sleeve **536** and waist portion **522** through which portion **548A**, **548B** exits, facilitating grasping of portion **548A**, **548B** when waist portion **522** is to be constricted, closed or reduced in size.

(80) When a person wishes to reduce or constrict the size of closable circumference **534** of waist portion **522**, the person may pull both of portions **548**. For example, the person may pull portion **548A** through length **550B** (as indicated by arrow **560A**), increasing the length of portion **548A** and reducing the length of portion **544A** extending about waist portion **522** and back to portion **542A**. The person may simultaneously pull portion **548B** through length **550A** (as indicated by arrow **560B**), increasing the length of portion **548B** and reducing the length of portion **544B** extending about waist portion **522** and back to portion **542A**. After the length of portions **544A**, **544B** and the size of the closable circumference of waist portion **522** have been established, the constriction of portions **550A** and **550B** retains the chosen size of the closable circumference of waist portion **522**. Outward forces exerted upon waist portion **522** and upon line portions **544A**, **544B** of lines **540A**, **540B** place an axial load on lengths **550A**, **550B** which constricts about and against portions **546A**, **546B**, respectively. The constriction of portions **550** about portions **546** frictionally locks portions **550** to portions **546** about the entire circumference or on all sides of portions **546**, inhibiting axial sliding movement of portions **546** within the hollow interior of constrict-able tubular lengths **550**.

(81) As shown by FIGS. 7 and 8, circumference closing system **430** provides garment **520** with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system **530** provides garment **520** with an easily adjustable waist portion **522** without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment **520**.

(82) Although each of the circumference closing systems **30**, **330**, **430** and **530** have been described as being incorporated into waist portions of various garments such as pants, shorts and skirts, each of the above described circumference closing systems **30**, **330**, **430** and **530** may additionally be incorporated into various other articles that have an adjustable mouth or circumference. For example, each of the above described circumference closing systems **30**, **330**, **430** and **530** may additionally be incorporated into any closed-loop waistband or other article having a circumference to be closed down, such as a waist opening of a jacket, a leg opening of pants, a sleeve opening, a cuff, a hem or the like. In addition to garments such as running shorts, sweatpants, swim shorts, skirts, jackets and the like, such closing systems may be incorporated into bags such as, backpacks, equipment bags, stuff sacks, travel bags, sling bags and other bags having a mouth or opening which can be secured in a closed state, but is openable for the insertion of objects.

(83) FIG. 13 illustrates an example bag **620** incorporating circumference closing system **530** described above. Bag **620** comprises a floor **622** and a mouth **624** having a circumference **534** which may be selectively opened and closed or constricted through the actuation of circumference closing system **530** described above. In such an implementation, circumference closing system **530** retains mouth **624** in a securely closed state. At the same time, pulling of exposed portion **552B** allows mouth **624** to be opened for the insertion or removal of articles.

(84) FIGS. 14-17 illustrate one example method for forming portions of circumference closing system **530**. As illustrated by FIG. 14, two separate lines **540A** and **540B** are provided. As illustrated by FIG. 15, a portion of line **540B** is passed through a constrict-able tubular portion **550A** of line **540A**. Likewise, a portion of line **540A** is passed through a constrict-able tubular portion **550B** of line **540B**. As shown by FIG. 15, in one implementation, the passing through of one line through to the interior portion of another line may be carried out using a tool **547**.

(85) As illustrated by FIGS. 16 and 17, end portions **542A** and **542B** are subsequently secured to the garment or other article through stitching or the like, directly or indirectly such as being

attached to a band, such as band **600**. As described above, the remaining portions of lines **540A** and **540B** may be contained within the sleeve of the garment or article, wherein at least one portion **552A** and/or **552B** is exposed to allow portions **552A** and/or **552B** to be pulled as indicated by arrows **658A** and **B**, facilitating opening or enlarging the circumference. Portions **548** may be grasped and pulled in the direction indicated by arrows **660** to constrict or close the circumference of the garment or article.

(86) Referring to FIG. **18** through **20**, example implementations of the present invention applied to an article of footwear **800** are illustrated. The article of footwear **800** can be a shoe, a sandal, a boot or any type of footwear. The article **800** of footwear includes an upper **802** and a tongue **804** defining an opening **806** for receiving the foot of a user, and a plurality of fixed eyelets **808** for routing of a lacing assembly **810**. The eyelets **808** are fixedly coupled to the upper **802** in a spaced apart manner about the perimeter of the opening **806**. In other implementations, the article of footwear can be formed without a tongue. In other implementations, the article of footwear can be formed with other structure or structures for securing the lacing such as, for example, one or more openings, one or more movable eyelets, one or more flexible eyelets, one or more anchors, one or more hooks, one or more latches and combinations thereof.

(87) The lacing assembly **810** includes a first lace **812**, a cord **814** and a floating eyelet **816**. The first lace **812** is an elongate flexible cord or line such as a lacing applied to existing articles of footwear. The first lace **812** has a first lace end **820** coupled to the upper **802** and a second lace end **822**. The cord **814** can be a generally circular braided elongate member including first and second cord ends **824** and **826** and a constricting tube **828** positioned between the first and second cord ends **824** and **826**. The constricting tube **828** can be a braided cord having hollow interior between first and second constricted tube openings **844** and **846**. In other implementations, the cord and the constricted tube can have other constructions. The first lace end **820** is fixedly coupled to the upper **802**, such as, through a stitching. In other implementations, the first lace end **820** can be fixedly coupled to the upper **802** in other manners. The second lace end **822** is then routed through a plurality of the fixed eyelets **808**, the floating eyelet **816**, another one of the eyelets **808**, the constricting tube **828** and an upper fixed eyelet **830** where the second lace end **822** is then available for grasping and pulling by the user. In other implementations, other numbers of eyelets or other fasteners can be used, the order or positioning of the eyelets (fixed and/or movable) and the constricting tube can be varied. The first cord end **824** is fixedly coupled to the upper **802** such as through a stitching. In other implementations, other means for fixing the first cord end to the upper can be used. The second cord end **826** is routed through the floating eyelet **816** and is anchored to the upper **802** and is looped through the upper fixed eyelet **830**. In other implementations, the cord **814** can take alternate paths through one or more eyelets or other fasteners and can be arranged between, before or after the constricting tube.

(88) The portion of the first lace **812** extending through the constricting tube **828** is referred to as a lockdown portion **832**. The portion of the first lace **812** between the first lace end **820** and the constricting tube **828** is referred to as a first lace portion **840**, and the portion of the first lace **812** extending between the constricting tube **828** and the second lace end **822** is the second lace portion **842**. The second lace portion **842** is an adjustable end portion of the first lace **812** in that its length is adjustable based upon which direction the first lace **812** is pulled through the constricting tube **828** by the user.

(89) The lacing assembly **810** provides a durable, reliable lacing for articles of footwear without requiring the use of one or more knots, or one or more fasteners or other hardware. The lacing assembly **810** uses the application of an axial load onto the braided cord **814** when the first lace **812** is tightened. The axial load is accomplished by fixing the first cord end **824** to a first location on the upper **802**, routing the cord **814** through the floating eyelet **816**, and then fixing the second cord **826** at a second location of the upper **802** typically closer to an upper portion of the shoe opening **806**. The first lace **812** can follow a traditional lacing path from the first lace end **820** fixed

to the upper **802** to the floating eyelet **816**, where the lace **812** extends to another of the eyelets **808** before entering the constricting tube **828** through the first opening **844**. The first lace **812** extends through the constricting tube **828** and exits the tube **828** at the second opening **846** where it extends through the upper fixed eyelet **830**. The orientation of the upper fixed eyelet **830** near or in line with the second opening **846** of the constricting tube **828** protects the second opening **846** of the constricting tube **828** from damage, wear or pulling out from pulling the second portion **842** of the first lace **812**.

(90) The lacing assembly **810** is then tightened by pulling on the second lace portion **842** of the first lace **812** (such as in the direction of arrow **848**). The pulling increases the load on the fixed upper eyelet **830** and the floating eyelet **816** creating an axial load on the braided cord **814** at the constricting tube **828** which causes the constricting tube to constrict, draw down or lock down onto the lockdown portion **832** of the lace **812** extending through the constricting tube **828**. The constricting, clamping or locking down of the constricting tube **828** of the cord **814** onto the lockdown portion **832** provides a secure fixed clamp or lock of the lace **812** with the cord **814** securing the lace **812** in place and keeping the article of footwear **800** securely on the user's foot. In one implementation, the article of footwear **800** may include a pouch **850** or pocket for storing the free second end **822** of the lace **812** after the lace system **810** has been tightened by pulling on the second portion **842** of the lace **812**.

(91) The lacing system **810** can be loosened by pulling on the first lace portion **840** (such as in the direction of arrow **852**). Pulling of the first portion **840** of the first lace **812** reduces the axial load on the constricting tube **828** thereby releasing or lessening of the constriction, clamping or locking-down of the portion **832** of the first lace **812** within the constricting tube **828** enabling the lace **812** to move relative to the constricting tube **828**.

(92) Referring to FIG. **19**, lacing system **910** is illustrated. The lacing system **910** is another implementation of the present invention and is substantially similar to the lacing system **810** except that lacing system **910** includes first and second cords **814A** and **814B** positioned on opposite sides of the opening **806** on the upper **802**, first and second floating eyelets **816A** and **816B**, and first and second upper eyelets **830A** and **830B**. Additionally, rather than being fixed to the upper as in system **810**, the first lace end **820** is not fixed or anchored to the upper, but it is routed through the eyelets **808**, the second constricting tube **828B** and the second upper eyelet **830B** to form a first adjustable end **820**. Accordingly, the lacing system **910** provides first and second adjustable ends **820** and **822**, which can be selectively positioned within the pocket **850**.

(93) The lacing system **910** is a double locking system due to the first and second constricting tubes **828A** and **828B**. The lacing system **910** can be tightened by pulling the first and second lace ends **820** and **822**, such as in the directions of **848A** and **848B**. Pulling on lace ends **820** and **822** increases the axial load or tension on the constricting tubes **828A** and **828B** causing the constricting tubes **828A** and **828B** to clamp down, constrict or lockdown on to the portions **832A** and **832B**, respectively, of the lace **812**. In order to loosen the lacing system **910**, a user can pull the portion of the lace **812** between the first and second constricting tubes **828A** and **828B**. The lace system **910** can further include a loosening ring **860** to facilitate the grasping and pulling of the lace **812** from the through the first openings **844A** and **844B** of the constricting tubes **828A** and **828B**. In other implementations, the loosening ring can take other shapes or configuration such as including a tab for facilitating the grasping of the loosening ring. The lacing system **910** provides a tight secure closure or tightening of the article of footwear **800** about the user's foot.

(94) FIG. **20** illustrates another implementation of the present invention. In particular, lacing system **1010** is illustrated. Lacing system **1010** is similar to lacing system **910** except that the lace **812** can be longer and the article of footwear **800** can be taller extending over more of the user's foot such as a boot or high top casual shoe. The article of footwear **800** can include a plurality of hooks **862** for facilitating the lacing of the lace **812** about the upper portion of the article of footwear **800**. In other implementations, other fastening mechanisms can be used to secure the

lacing through the upper portion of the footwear article, such as, for example, eyelets, openings, loops, other fasteners, and combinations thereof.

(95) Although the present disclosure has been described with reference to example implementations, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the claimed subject matter. For example, although different example implementations may have been described as including one or more features providing one or more benefits, it is contemplated that the described features may be interchanged with one another or alternatively be combined with one another in the described example implementations or in other alternative implementations. Because the technology of the present disclosure is relatively complex, not all changes in the technology are foreseeable. The present disclosure described with reference to the example implementations and set forth in the following claims is manifestly intended to be as broad as possible. For example, unless specifically otherwise noted, the claims reciting a single particular element also encompass a plurality of such particular elements. The terms “first”, “second”, “third” and so on in the claims merely distinguish different elements and, unless otherwise stated, are not to be specifically associated with a particular order or particular numbering of elements in the disclosure.

Claims

1. An article comprising: a flexible perimeter defining a closable circumference; and a circumference closing system comprising: a first line, the first line comprising: a first portion of the first line secured to the closable circumference at a first location; a second portion of the first line having a first constrict-able tubular length with a first entry location and a first exit location; a third portion of the first line extending from the second portion along and in a first direction about the closable circumference; and a second line, the second line comprising: a first portion of the second line secured to the circumference at a second location; a second portion of the second line having a second constrict-able tubular length with a second entry location and a second exit location; a third portion of the second line extending from the second portion along and in a second direction, opposite the first direction, about the closable circumference, wherein the first line further comprises: a fourth portion of the first line extending from the third portion into the second entry location and through the second constrict-able tubular length; and a fifth portion of the first line exiting the second constrict-able tubular length at the second exit location, and wherein the second line further comprises: a fourth portion of the second line extending from the third portion of the second line into the first entry location and through the first constrict-able tubular length; and a fifth portion of the second line exiting the first constrict-able tubular length at the first exit location of the first line.
2. The article of claim 1, wherein the fifth portion of the first line and the fifth portion of the second line are secured to the article.
3. The article of claim 1, wherein an entire length of the first line and an entire length of the second line are each a constrict-able tube.
4. The article of claim 1, wherein each constrict-able tube comprises a braided tube.
5. The article of claim 1, wherein the first constrict-able tubular length and the second constrict-able tubular length each comprise a braided constrict-able tubular length.
6. The article of claim 1, wherein the article comprises a sleeve comprising first and second gaps and first and second openings, wherein the first constrict-able tubular length extends within the sleeve between the first location and the first gap, wherein the second constrict-able tubular length extends within the sleeve between the second location and the second gap, wherein the third portion of the first line extends across the second gap and wherein the third portion of the second line extends across the first gap.
7. A bag comprising: a floor; a mouth providing access to an interior of the bag; and a mouth

closing system comprising: a first line, the first line comprising: a first portion of the first line secured to the mouth at a first location; a second portion of the first line having a first constrict-able tubular length with a first entry location and a first exit location; a third portion of the first line extending from the second portion along and in a first direction about the mouth; and a second line, the second line comprising: a first portion of the second line secured to the mouth at a second location; a second portion of the second line having a second constrict-able tubular length with a second entry location and a second exit location; a third portion of the second line extending from the second portion along and in a second direction, opposite the first direction, about the mouth, wherein the first line further comprises: a fourth portion of the first line extending from the third portion into the second entry location and through the second constrict-able tubular length; and a fifth portion of the first line exiting the second constrict-able tubular length at the second exit location, and wherein the second line further comprises: a fourth portion of the second line extending from the third portion of the second line into the first entry location and through the first constrict-able tubular length; and a fifth portion of the second line exiting the first constrict-able tubular length at the first exit location of the first line.

8. The bag of claim 7, wherein the fifth portion of the first line and the fifth portion of the second line are secured to the bag.

9. The bag of claim 7, wherein an entire length of the first line and an entire length of the second line are each a constrict-able tube.

10. The bag of claim 7, wherein each constrict-able tube comprises a braided tube.

11. The bag of claim 7, wherein the first constrict-able tubular length and the second constrict-able tubular length each comprise a braided constrict-able tubular length.

12. The bag of claim 7, wherein the bag comprises a sleeve comprising first and second gaps and first and second openings, wherein the first constrict-able tubular length extends within the sleeve between the first location and the first gap, wherein the second constrict-able tubular length extends within the sleeve between the second location and the second gap, wherein the third portion of the first line extends across the second gap and wherein the third portion of the second line extends across the first gap.
